STAT 6206: Bayesian Statistics with Applications in Machine Learning Assignment 1

Academic year 25/26, second term

Due date: February 11, 2026

1. Galenshore Distribution: An unknown quantity Y has a Galenshore(a, θ) distribution if its density is

$$p(y \mid \theta) = \frac{2}{\Gamma(a)}\, \theta^{2a}\, y^{2a-1} \exp\big(-\theta^2 y^2\big),\ y > 0,\ \theta > 0,\ a > 0.$$

Assume for now that a is known. For this density,

$$\mathbb{E}[Y] = \frac{\Gamma(a + \frac{1}{2})}{\theta\, \Gamma(a)},\ \mathbb{E}[Y^2] = \frac{a}{\theta^2}.$$

(a) Identify a class of conjugate prior densities for θ. (Hint: Let $\phi = \theta^2$)

(b) Let $Y_1, \ldots, Y_n \overset{iid}{\sim}$ Galenshore(a, θ). Find the posterior distribution of θ given $Y_1, \ldots, Y_n$, using a prior from your conjugate class.

(c) Determine $\mathbb{E}[\theta \mid y_1, \ldots, y_n]$ (Hint: If $X \sim Gamma(k, r)$, then $E[X^s] = r^{-k} \cdot \frac{\Gamma(k+s)}{\Gamma(k)}, (s > -k)$).

2. MLE vs Bayesian Estimator: Let $Y_1, \ldots, Y_n \overset{iid}{\sim}$ Poisson(λ) with $\lambda > 0$. Denote $y_{1:n} = (y_1, \ldots, y_n)$ and $S = \sum_{i=1}^n y_i$.

(a) Write down the likelihood $L(\lambda \mid y_{1:n})$ and the log-likelihood $\ell(\lambda)$. Derive the maximum likelihood estimator (MLE) $\hat{\lambda}_{\text{MLE}}$.

(b) Assume a conjugate Gamma prior in *shape–rate* parameterization:

$$\lambda \sim \text{Gamma}(a, b),\ a > 0,\ b > 0.$$

Derive the posterior distribution $p(\lambda \mid y_{1:n})$ and identify its shape and rate parameters. Compute the posterior mean and the posterior variance.

3. Normal model with known variance: In a study examining the weights of a certain breed of dogs, researchers collect a sample of n dogs and measure their weights. Assume

$$Y_i \mid \mu \overset{\text{iid}}{\sim} N(\mu, \sigma^2),\ i = 1, \ldots, n,$$

where the variance σ^2 is known.

(a) Assume a prior distribution for the mean: $\mu \sim N(\mu_0, \tau^2)$, where μ_0 and τ^2 are known, compute the posterior distribution $p(\mu \mid y_1, \ldots, y_n)$.

(b) Derive the posterior predictive distribution $p(\tilde{Y} \mid y_1, \ldots, y_n)$ for the weight $\tilde{Y}$ of a new dog.

(c) Calculate the posterior predictive mean of $\tilde{Y}$ and construct a 95% credible interval for $\tilde{Y}$.

4. Binomial GLMM: Consider a dataset where the response variable y_{it} for the i-th observation at time t follows a Binomial distribution with the model form:

$$g(p_{it}) = \text{logit}(p_{it}) = \beta_0 + \beta_1 x_{it1} + \cdots + \beta_p x_{itp} + u_i,\ i = 1, \ldots, n,\ t = 1, \ldots, T,$$

where $Y_{it} \mid p_{it} \sim \text{Binomial}(m_{it}, p_{it})$, and:

- p_{it} is the success probability for the i-th observation at time t,
- $g(\cdot)$ is the link function,
- β_j are the fixed effect parameters,
- u_i is the random effect, assumed to follow $u_i \sim \mathcal{N}(0, \sigma_u^2)$.

(a) Specify the prior distribution for all unknowns in the model.

(b) Derive the full conditional distribution for these unknowns.